JEREMIAH NYAGAH NATIONAL POLYTECHNIC

DEPARTMENT: ELECTRICAL AND ELECTRONICS ENGINEERING

LEVEL 6

SUBJECT: ELECTRICAL POWER LINES

CAT 1

PERIOD: 2 HRS

Instruction:

a) Attempt **ALL** the **QUESTIONS**
b) Any form of cheating is not allowed
c) Phone are not allowed in the exam room

QUESTIONS

1. State any five elements of a transmission line. (5 marks)
2. Differentiate a long transmission line, medium transmission line and short transmission line. (3 marks)
3. Explain a transmission Line. (2 marks)
4. Give four types of line supports. (4 marks)
5. State three materials used for making line supports. (3 marks)
6. State Five types of transmission line conductors. (5 marks)
7. Describe factors affecting the sag. (5 marks)
8. State four types of electrical joints. (4 marks)
9. State three type of electrical load. (3 marks)

10. What do you understand by the term “transmission line termination.” (2 marks)
11. Describe three types of transmission line termination. (3 marks)
12. What is the importance of termination? (1 mark)
13. Differentiate between AC and DC transmission (2 marks)
14. State the advantages and disadvantages of AC transmission (5 marks)
15. State the advantages and disadvantages of DC transmission? (5 marks)
16. State testing in line transmission. (1 mark)
17. State FOUR systems of power transmission. (4 marks)
18. State FOUR important components of an overhead transmission line. (4 marks)
19. Highlight FOUR advantages of pin-type insulator used in power transmission lines.
20. (4 marks)
21. Explain TWO purposes of suspension insulators as used in high voltage power transmission. (4 marks)
22. Outline THREE methods of improving string efficiency as used in design of transmission lines. (6 marks)
23. Explain TWO methods of reducing corona effect in an overhead transmission line.
24. (4 marks)
25. Define the term “sag” as used in overhead lines. (2 marks)
26. Outline TWO factors which affect the sag of overhead transmission lines. (2 marks)
27. State THREE classifications of overhead transmission lines citing their voltage levels. (6 marks)
28. Outline FOUR properties of conductor material used for transmission and distribution of electric power. (4 marks)